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India's 1st 5.56 km Free-Space Quantum Key Distribution Link Demonstrated



Source: PIB

Why in News?

India has successfully demonstrated its first **5.56 km Free-Space Quantum Key Distribution (QKD)** link. The field trial established a secure quantum communication channel between **BISAG-N** (Bhaskaracharya National Institute for Space Applications and Geo-informatics) and **IIT Gandhinagar**. The demonstration was executed in collaboration with **QNu Labs**, an Indian deep-tech startup.

What is Free-Space Quantum Key Distribution (QKD)?

- **Quantum Key Distribution (QKD):** It is a secure communication method that uses the **quantum properties of photons** to exchange cryptographic keys between two distant parties.
 - It **does not transmit the actual message through quantum states**. Instead, QKD securely establishes an **encryption key**, which is then used by conventional systems to **encrypt and decrypt the actual data**.
 - **Free-Space QKD** is a quantum-secure communication technology that transmits encryption keys through **laser beams in open air**, eliminating the need for physical fibre-optic cables.
 - QKD relies on **Heisenberg's Uncertainty Principle and the No-Cloning Theorem**.
- **Heisenberg's Uncertainty Principle:** According to quantum mechanics, measuring a quantum system inherently alters it. If an eavesdropper tries to intercept the photons, the quantum state collapses, instantly alerting the communicating parties of a breach.
- **No-Cloning Theorem:** It states that it is **physically impossible to create a perfect**, identical copy of an unknown quantum state, preventing hackers from secretly duplicating the transmitted keys.
- **Eavesdropper Detection:** QKD does not transmit the actual message. Instead, it generates a secure cryptographic key used to encrypt the data.
 - Any attempt to intercept or measure the quantum states disturbs them and introduces detectable errors, allowing the users to detect eavesdropping and discard the compromised key.
- **Atmospheric Challenges:** Unlike fibre-based QKD, free-space QKD **requires direct line-of-sight between transmitter and receiver**. Precise alignment is necessary, while atmospheric turbulence

and ambient light pollution can distort or weaken the optical signal.

India's Breakthrough

The recent 5.56 km trial is a major leap from earlier short-distance tests (like the 300m test in 2021). Key highlights include:

- **Quantum Bit Error Rate (QBER):** Measures the proportion of quantum bits received incorrectly. Lower QBER means a more reliable and secure quantum channel. In the 5.56 km trial, QBER remained below 5%.
- **Advanced Tracking:** The trial successfully utilized QNu Labs’ **precise Pointing, Acquisition, and Tracking (PAT) system** to perfectly align the laser beam across open air over the entire distance.
- **Two-Layered Security Architecture:** The demonstration combined QNu Labs' **hardware (Armos) with BISAG-N’s software (Vedic Kavach)**.
 - This created a hybrid setup where **QKD** was layered with **Post-Quantum Cryptography (PQC)** (*encryption algorithms designed to remain secure against quantum computers*) .
 - If the physical atmospheric link drops temporarily due to weather, the mathematical PQC software continues to protect the data.
- **Significance for India:**
 - **Precursor to Satellite QKD:** Free-space terrestrial QKD is a key stepping stone toward satellite-based quantum communication, as both rely on transmitting quantum signals through the atmosphere between line-of-sight optical terminals.
 - **National Security:** As quantum computers become advanced enough to break classical encryption algorithms, this technology provides a "**quantum-safe**" **shield for defence, strategic infrastructure, and financial networks against future cyber threats.**
 - **National Quantum Mission Target:** India’s **National Quantum Mission (NQM)** aims to develop satellite-based secure quantum communication between ground stations separated by up to 2,000 km, along with a 2,000-km inter-city QKD network using optical fibre.

Free-Space vs. Fibre-Based QKD

Feature	Fibre-Based QKD	Free-Space QKD
Transmission Medium	Fibre-optic cables	Open air, atmosphere, or space
Distance	A few hundred km without trusted repeaters due to signal loss	Can cover thousands of km, especially via satellites
Environmental Impact	More stable and protected from weather	Affected by fog, rain, turbulence, and daylight
Alignment	Fixed connection; no active tracking needed	Requires precise Line-of-Sight (LoS) and active pointing and tracking

Note : Earlier, under the NQM, startup QNu Labs demonstrated India's first 500 km QKD network using existing optical-fibre infrastructure, strengthening quantum-secure communication and cyber defence.

- On a global scale, China launched the **world's first quantum-communication satellite, Micius, in 2016**. It operates in **low-Earth orbit** at around 500 km. China later launched Jinan-1 in 2022, which demonstrated real-time QKD with portable ground stations.

Click here to Read: India Demonstrates 500 km Quantum Key Distribution Network

Frequently Asked Questions (FAQs)

1. What is Quantum Key Distribution (QKD)?

QKD is a secure method of **exchanging cryptographic keys using quantum properties of photons**, enabling detection of interception attempts.

2. What is Free-Space QKD?

Free-Space QKD transmits quantum encryption keys through **laser beams in open air**, without requiring physical fibre-optic cables.

3. What was significant about India's 5.56 km QKD trial?

It demonstrated a **5.56 km secure free-space quantum communication link** between **BISAG-N and IIT Gandhinagar**, supporting India's future satellite-QKD capabilities.

4. Why is Post-Quantum Cryptography (PQC) important?

PQC uses **quantum-resistant mathematical algorithms** to protect conventional data from future attacks by powerful quantum computers.

5. What are the major challenges of Free-Space QKD?

Its deployment is affected by **atmospheric turbulence, fog, rain and daylight**, while requiring highly precise **Line-of-Sight and Pointing, Acquisition and Tracking (PAT)** systems.

[Watch Video on YouTube: [▶ https://www.youtube.com/embed/doZ3q13v3UE](https://www.youtube.com/embed/doZ3q13v3UE)]

UPSC Civil Services Examination Previous Year Question (PYQ)

Q. Which one of the following is the context in which the term "qubit" is mentioned? (2022)

- (a) Cloud Services
- (b) Quantum Computing
- (c) Visible Light Communication Technologies
- (d) Wireless Communication Technologies

Ans: (b)